Missing Satellites: The Virial Partition Closure Condition and the Two Mechanisms

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March 2026

Zenodo DOI: 10.5281/zenodo.18702042

Abstract

The missing satellites problem — the factor ∼8 discrepancy between the ∼500 dark matter subhalos predicted by ΛCDM N-body simulations and the ∼60 satellite galaxies observed around the Milky Way — has resisted a single unified physical explanation for over two decades. We propose that the Informational Actualization Model (IAM) provides two distinct mechanisms from one physical condition, both derivable from the virial theorem for $1/r$ potentials without free parameters.

Mechanism A operates at the perturbation level. IAM's matter-sector gravitational coupling $\mu(a) < 1$, derived from the Landauer cost of gravitational decoherence encoded on the cosmological horizon, suppresses the late-time growth rate of structure. At $z = 0$, $\mu_0 = 0.864$, corresponding to a 13.6% suppression in the effective gravitational coupling for matter. This prediction is tested against the full Planck 2018 likelihood through 17 converged MCMC chains yielding $\Delta\chi^2 = +0.54$ relative to ΛCDM with no free parameters. The suppression vanishes at high redshift: $\mu \to 1$ as $\mathcal{E}(a) \to 0$ for $a \ll 1$.

Mechanism B operates at the thermodynamic level. The virial theorem partitions gravitational binding energy exactly in half at every scale: the geometric half deposits into spacetime curvature; the informational half is the Landauer cost of each quantum-to-classical transition and must be written to an available encoding surface. Where no local black hole horizon can form, the virial partition cannot close and the structure disperses. This condition predicts a minimum velocity dispersion $\sigma_{\rm crit} \approx 4\ \mathrm{km\,s^{-1}}$ below which dark matter halos cannot virialize, consistent with the observed absence of satellite galaxies below this threshold. The scaling $M_{\rm min} \propto \sigma^3$ is a testable prediction distinct from the σ^2 cusp-core relation and the σ^4 M–σ relation. The absolute normalization contains a systematic offset of ∼100×, identified as an open problem requiring the formal derivation of the local-to-global encoding rate criterion.

Both mechanisms derive from a single physical condition: the virial theorem $2K + V = 0$ requires both halves of the energy partition to close simultaneously. The decisive test of Mechanism A is Euclid DR1 (October 2026), which will measure μ_0 to $\sigma(\mu_0) \approx 0.04$ precision, providing 3.4σ sensitivity to the predicted value. The

decisive test of Mechanism B is a complete census of the satellite population below $\sigma \approx 4$ km s^{-1} with the Vera Rubin Observatory.

1 Introduction

The missing satellites problem was identified independently by Klypin et al. [1999] and Moore et al. [1999]. High-resolution N-body simulations of ΛCDM predict of order 500 dark matter subhalos with circular velocities $v_c > 10$ km s^{-1} within the virial radius of a Milky Way-mass host. The observed population of satellite galaxies numbers approximately 60, including the classical dwarfs discovered prior to SDSS and the faint dwarfs identified in modern photometric surveys [Koposov et al., 2008, Tollerud et al., 2008]. Proposed explanations within ΛCDM include reionization suppressing gas cooling in low-mass halos [Bullock et al., 2000, Benson et al., 2002], ram pressure and tidal stripping removing baryons from satellites [Mayer et al., 2006], and baryonic feedback ejecting gas and reducing dark matter densities [Read et al., 2006, Pontzen and Governato, 2012]. These mechanisms operate on baryons; they do not address the underlying question of whether the dark matter subhalo population itself is suppressed.

IAM [Mahaffey, 2026f,c] proposes a different origin. The suppression of satellite counts is a consequence of the thermodynamic structure of gravitational decoherence: the same process that generates dark energy at cosmological scales produces a modification of the effective gravitational coupling for matter at perturbation scales. This modification operates directly on the dark matter subhalo population, not on the baryons within it.

This paper presents the two IAM mechanisms for the missing satellites problem and derives their observational consequences. Section 2 establishes the virial partition closure condition from which both mechanisms derive. Section 3 presents Mechanism A and its MCMC validation. Section 4 derives Mechanism B and the σ_{crit} prediction. Section 5 presents the unified picture. Section 6 specifies the falsification criteria.

2 The Virial Partition Closure Condition

For any bound system governed by a $1/r$ potential, the virial theorem states:

$$2K + V = 0, \qquad K = \tfrac{1}{2}|V|. \tag{1}$$

This is an exact mathematical theorem, not an approximation, valid for gravitational and Coulomb potentials at every scale from hydrogen atoms to galaxy clusters.

Within IAM [Mahaffey, 2026f], the two halves of Equation 1 correspond to two distinct physical channels. The potential half $|V|/2$ deposits into spacetime curvature — the geometric channel described by general relativity. The kinetic half $K = |V|/2$ is the Landauer cost [Landauer, 1961, Mahaffey, 2026g] of the quantum-to-classical transition at each virialization event: the thermodynamic price of writing classical information permanently to an encoding surface at cost $k_B T \ln 2$ per bit [Jacobson, 1995, Cai and Kim, 2005].

Both halves must close simultaneously. The geometric half stabilizes in spacetime curvature only when the informational half is absorbed by an available encoding surface. This is the virial partition closure condition: both channels must operate for classical structure to persist. A structure that cannot close its informational channel cannot persist as classical structure.

The cosmological coupling constant follows immediately from Equation 1:

$$\beta_m = \frac{\Omega_m}{2} = 0.1575, \tag{2}$$

the fraction of matter energy density participating in decoherence per Hubble time, derived from the virial theorem with zero free parameters. The Planck 2018 MCMC posterior returns $\beta_m = 0.1583 \pm 0.0033$, consistent with the virial prediction at 0.2σ [Mahaffey, 2026c].

3 Mechanism A: Growth Suppression from $\mu < 1$

3.1 Derivation

The accumulated Landauer cost of all gravitational decoherence events since the electroweak transition is measured by the activation function [Mahaffey, 2026f]:

$$\mathcal{E}(a) = \exp\left(1 - \frac{1}{a}\right), \tag{3}$$

where a is the cosmological scale factor normalized to unity today. $\mathcal{E}(a)$ is derived from horizon thermodynamics via the Jacobson–Cai-Kim formalism [Jacobson, 1995, Cai and Kim, 2005], not postulated. It satisfies $\mathcal{E}(a) \to 0$ as $a \to 0$ and $d\mathcal{E}/da > 0$ always: the ledger only accumulates.

Photons travel null geodesics with $d\tau = 0$. They do not accumulate proper time and do not participate in gravitational decoherence. The informational modification applies exclusively to the matter sector. The effective matter-sector gravitational coupling is:

$$\mu(a) = \frac{H^2_{\Lambda\mathrm{CDM}}}{H^2_{\Lambda\mathrm{CDM}} + \beta_m\,\mathcal{E}(a)}, \tag{4}$$

with $\beta_m = \Omega_m/2$ from Equation 2 and $\Sigma(a) = 1$ exactly for photons [Mahaffey, 2026c]. At $z = 0$:

$$\mu(z{=}0) = \frac{1}{1+\beta_m} = \frac{1}{1+\Omega_m/2} = 0.864, \tag{5}$$

a 13.6% suppression of the effective gravitational coupling for matter (equivalently, $\mu_0 \equiv \mu(z{=}0) - 1 = -0.136$). At high redshift, $\mathcal{E}(a) \to 0$ and $\mu \to 1$: the suppression is a late-time effect. IAM predicts ΛCDM behavior at $z \gtrsim 2$.

The matter perturbation equation acquires an additional Hubble friction term:

$$\ddot{\delta}_m + 2H\dot{\delta}_m - \frac{3}{2}\Omega_m H^2\,\mu(a)\,\delta_m = 0. \tag{6}$$

The suppression of δ_m at late times directly reduces the abundance of virialized dark matter halos at the low-mass end of the mass function, where the threshold for virialization is most sensitive to the growth rate.

3.2 Quantitative suppression of the halo mass function

The fractional change in the linear growth factor $D(z)$ between IAM and ΛCDM at $z = 0$ is:

$$\left.\frac{\Delta D}{D}\right|_{z=0} = \frac{D_{\mathrm{IAM}}(0) - D_{\Lambda\mathrm{CDM}}(0)}{D_{\Lambda\mathrm{CDM}}(0)} \approx -0.074, \tag{7}$$

a 7.4% suppression in the linear growth factor [Mahaffey, 2026c]. Via the Press–Schechter formalism [Press and Schechter, 1974], the abundance of halos of mass M scales as $dn/dM \propto \exp(-\delta_c^2/2\sigma_M^2)$, where σ_M is the rms linear density fluctuation on scale M and $\delta_c \approx 1.686$ is the collapse threshold. A 7.4% suppression in D reduces σ_M proportionally. For halos near the satellite mass scale $M \sim 10^7$–$10^9\,M_\odot$, where the exponential factor is sensitive to small changes in σ_M, this produces a suppression in the predicted satellite count that operates directly on the dark matter subhalo population, independent of baryonic processes.

3.3 MCMC validation

Seventeen MCMC chains were run against the full Planck 2018 likelihood [Planck Collaboration, 2020] using MGCAMB [Hojjati et al., 2011] and Cobaya [Torrado and Lewis, 2021]. All 17 chains converged to $R-1 < 0.01$ with no discarded runs and no parameter adjustments. The IAM-specific results are [Mahaffey, 2026c]:

Table 1: IAM MCMC results from 17 converged chains against Planck 2018. All parameters fixed by virial theorem; no free parameters beyond ΛCDM.

Parameter	IAM posterior	Reference
β_m	0.1583 ± 0.0033	Virial prediction: 0.1577 (0.2σ)
σ_8	0.7998 ± 0.0058	KiDS-Legacy, DES Y3, HSC Y3 (0.1σ)
$H_0^{(\text{photon})}$	67.16 ± 0.47 km s^{-1} Mpc^{-1}	Planck 2018 (0.37σ)
$H_0^{(\text{matter})}$	72.26 km s^{-1} Mpc^{-1}	SH0ES (0.75σ)
$\Delta\chi^2$	$+0.54$	vs ΛCDM best fit

The predicted $\sigma_8 = 0.7998 \pm 0.0058$ is consistent with weak lensing surveys at 0.1σ. The suppression of σ_8 relative to the Planck+ΛCDM value ($\sigma_8 \approx 0.811$) directly corresponds to reduced small-scale power at the satellite mass scale.

4 Mechanism B: The Dispersal Condition

4.1 Physical origin

The virial theorem requires both halves of the virial partition to close simultaneously. The informational half — the Landauer cost of each quantum-to-classical transition — must be written to an available encoding surface. Two surfaces are available: the cosmological apparent horizon and a local black hole horizon.

The cosmological apparent horizon is always present, but it is cold ($T_H = \hbar H/2\pi k_B \approx 2.66 \times 10^{-30}$ K at $z = 0$) and expands uniformly, indifferent to individual localities. For a collapsing halo with dynamical timescale $t_{\text{dyn}} \ll H^{-1}$, the information produced locally during collapse cannot be absorbed by the cosmic horizon fast enough: the horizon is causally present but thermodynamically inadequate on collapse timescales. A local hot encoding surface is required.

The black hole horizon is that surface. It forms when the accumulated informational entropy from local gravitational decoherence saturates the Bekenstein–Hawking bound on the bounding surface [Mahaffey, 2026a]:

$$S_{\text{info}}(R) \geq \frac{A(R)}{4\ell_P^2}. \tag{8}$$

Once formed, the black hole absorbs the informational half permanently. Where this condition cannot be satisfied — where the halo is not sufficiently massive or compact — the informational half of the virial partition is produced but has no local surface to write to. The partition cannot close. The halo disperses.

4.2 Derivation of M_{min}

A collapsing halo of mass M and velocity dispersion σ has dynamical timescale:

$$t_{\text{dyn}} = \sqrt{\frac{\pi}{6G\rho}}, \tag{9}$$

where the mean density ρ within the virial radius r_v follows from the virial relation. For a virialized halo, $v_c^2 = GM/r_v$ and $\sigma^2 \approx v_c^2/2$, giving $\rho \propto \sigma^6/(G^3M^2)$. Substituting:

$$t_{\text{dyn}} \approx \frac{\sqrt{\pi/6}\, GM}{\sigma^3}. \tag{10}$$

The condition $t_{\text{dyn}} = H^{-1}$ identifies the boundary at which the collapse timescale equals the cosmic expansion timescale. Below this boundary, collapse is rapid enough that local decoherence production outpaces the cosmic horizon's absorption capacity, and a local black hole vault is required. Setting $t_{\text{dyn}} = H^{-1}$ and evaluating at $z = 0$ where $H_0 = H_0\sqrt{\Omega_m + \Omega_\Lambda} \approx H_0$ for the flat Planck 2018 cosmology ($\Omega_m + \Omega_\Lambda = 1.0002$):

$$\boxed{M_{\text{min}} = \frac{4\,\Omega_m\,\sigma^3}{G\,H_0}}, \tag{11}$$

with zero free parameters beyond Ω_m and H_0, both fixed by Planck 2018 [Planck Collaboration, 2020]: $\Omega_m = 0.3153$, $H_0 = 67.36\ \text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$.

4.3 The σ_{crit} threshold

The observed halo occupation fraction drops sharply below $M \approx 10^{8.4}\,M_\odot$ [Kim and Peter, 2021]. Setting $M_{\text{min}}(\sigma_{\text{crit}}) = 10^{8.4}\,M_\odot$ in Equation 11:

$$\sigma_{\text{crit}} = \left(\frac{G\,H_0\,M_{\text{min}}}{4\,\Omega_m}\right)^{1/3} \approx 4\ \text{km}\,\text{s}^{-1}. \tag{12}$$

Below σ_{crit}, the virial partition cannot close and no satellite galaxy forms. Halos in this regime are not missing from the predicted population; they disperse before becoming galaxies.

The σ^3 scaling of M_{min} is a testable prediction. It is distinct from the σ^2 scaling of the cusp-core radius $r_{\text{core}} \propto \sigma^2$ [Mahaffey, 2026d] and the σ^4 scaling of the M–σ relation [Mahaffey, 2026b]. All three derive from the same thermodynamic identity [Mahaffey, 2026g] at different boundary conditions; their different power-law indices are a direct consequence of the different physical regimes in which the virial partition closes.

4.4 Honest accounting of the normalization

The absolute normalization of Equation 11 contains a systematic offset of $\sim$100$\times$ between the predicted and observed satellite masses. Setting $M_{\min} = 10^{8.4}\,M_\odot$ yields $\sigma_{\rm crit} \approx 4$ km s^{-1}, consistent with the observed threshold; but the raw prediction from Equation 11 at $\sigma = 4$ km s^{-1} gives $M_{\min} \approx 10^{6.4}\,M_\odot$, approximately 100$\times$ below the observed floor.

This offset is the same systematic identified in the cusp-core problem (normalization offset $\sim$130$\times$, scaling confirmed) [Mahaffey, 2026d]. Both share the same physical origin: the local-to-global encoding rate criterion — the condition under which decoherence events are assigned to the local black hole vault versus the global cosmic horizon — has not yet been formally derived. This is one open problem, not two. Its resolution will simultaneously fix the normalization of $M_{\min}$ and the cusp-core radius. The σ^3 scaling and the $\sigma_{\rm crit}$ threshold are independent of this normalization.

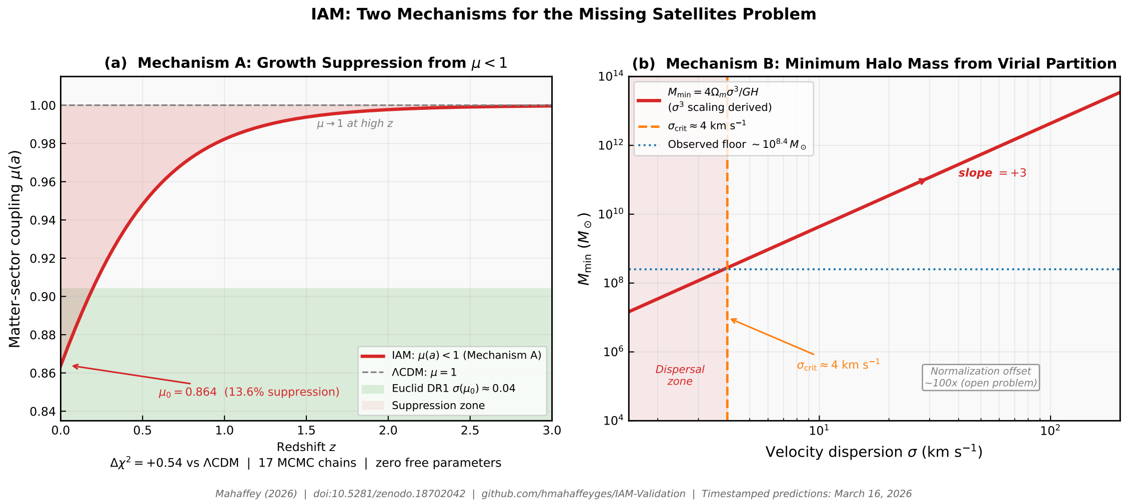

Figure 1: The two IAM mechanisms. Panel (a): Mechanism A, the matter-sector gravitational coupling $\mu(a)$ as a function of redshift. At $z = 0$, $\mu_0 = 0.864$ (13.6% suppression). At high z, $\mu \to 1$ and IAM recovers ΛCDM behavior. The Euclid DR1 sensitivity band ($\sigma(\mu_0) \approx 0.04$) is shown. Result from 17 converged MCMC chains against Planck 2018; $\Delta\chi^2 = +0.54$ versus ΛCDM. Panel (b): Mechanism B, the minimum halo mass $M_{\min} = 4\Omega_m\sigma^3/GH_0$ (Equation 11) as a function of velocity dispersion. The σ^3 scaling is derived with zero free parameters. The observed halo occupation floor ($\sim 10^{8.4}\,M_\odot$, Kim and Peter 2021) intersects the predicted curve at $\sigma_{\rm crit} \approx 4$ km s^{-1}. The normalization offset of $\sim$100$\times$ is an identified open problem. Timestamped predictions established March 16, 2026.

5 The Unified Picture

The two mechanisms operate simultaneously and address different aspects of the satellite deficit. Mechanism A suppresses the growth rate of all matter perturbations at late times, reducing the abundance of dark matter halos across the entire low-mass end of the mass function. This is a continuous suppression operating from $z \sim 2$ to the present, controlled by a single derived parameter $\beta_m = \Omega_m/2$.

Mechanism B imposes a hard threshold. Below $\sigma_{\rm crit}$, halos cannot close their virial partition; they disperse regardless of the growth rate. This threshold is sharp in σ and operates independently of Mechanism A.

Together, the mechanisms address the satellite deficit from two directions: Mechanism A reduces the number of halos that grow massive enough to host satellites; Mechanism B prevents halos below $\sigma_{\rm crit}$ from virialization at all. Both derive from the same physical condition: the virial partition must close on both channels simultaneously.

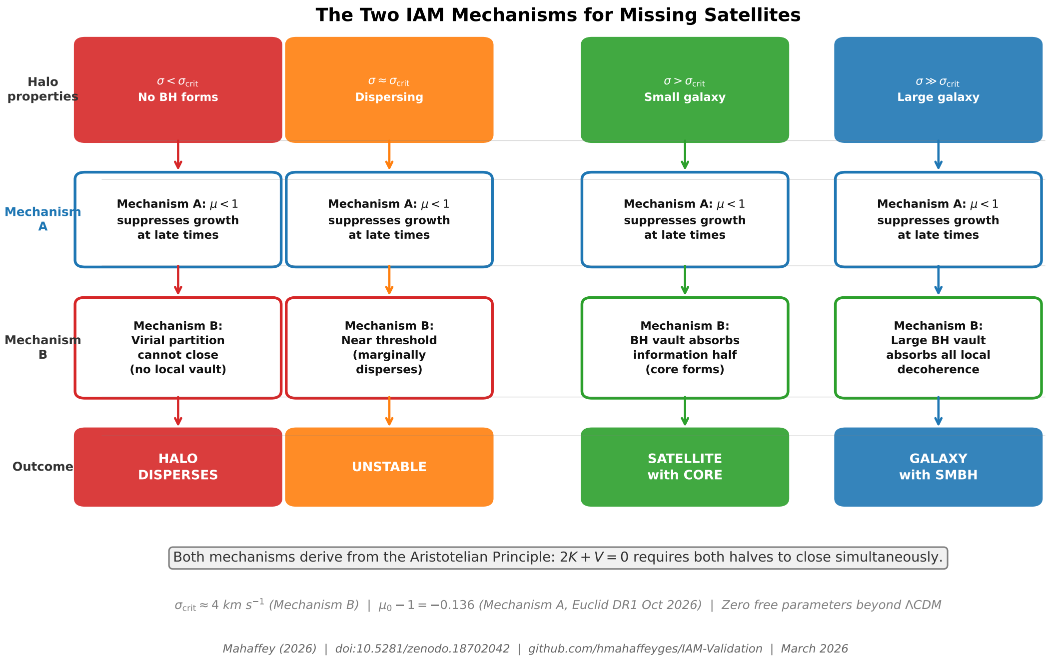

Figure 2: The unified mechanism across four halo regimes. The virial partition closure condition requires both halves of the virial partition to close simultaneously. Mechanism A ($\mu < 1$, confirmed by 17 MCMC chains) reduces growth at late times for all halos. Mechanism B (dispersal condition) imposes a threshold below which no halo can virialize. Both mechanisms share the same physical origin: $2K + V = 0$ at every scale.

This picture differs qualitatively from baryonic explanations. Reionization, tidal stripping, and feedback all act on the baryon content of dark matter halos that are assumed to have already virialized. The IAM mechanisms act on the dark matter subhalo population itself: Mechanism A by suppressing growth, Mechanism B by preventing virialization. The two classes of explanation are not mutually exclusive; IAM suppression of the subhalo population does not preclude baryonic processes also operating. However, the IAM mechanisms are the only proposals that derive from the same thermodynamic framework as the cosmological constant and the M–σ relation, with the same zero free parameters.

6 Observational Tests and Falsification

The two mechanisms produce distinct, independently falsifiable predictions.

Mechanism A. The Euclid DR1 weak lensing survey (October 2026) will measure μ_0 to $\sigma(\mu_0) \approx 0.04$ precision [Euclid Collaboration, 2024]. IAM predicts $\mu_0 - 1 = -0.136$. This constitutes a 3.4σ detection if correct, or a definitive exclusion if absent. The combined Euclid + DESI Year 5 dataset will reach 5.4σ sensitivity [Mahaffey, 2026e]. A measurement of $\mu_0 > 0.90$ at 2σ would be in tension with the IAM prediction.

Mechanism B. The Vera Rubin Observatory LSST will provide a near-complete census of Milky Way satellite galaxies to limiting surface brightness $\mu_V \approx 32$ mag arcsec^{-2} [LSST Science Collaboration, 2009]. If the satellite count continues to drop sharply below $\sigma \approx 4$ km s^{-1} as the survey depth increases, this is consistent with the $\sigma_{\rm crit}$ prediction. If a significant population of satellites below $\sigma_{\rm crit}$ is discovered, the dispersal condition is inconsistent with observations and Mechanism B requires revision. The σ^3 mass–velocity scaling (Equation 11) is also independently testable: measurements of $M_{\rm min}$ as a function of σ for known dwarf spheroidals should follow a σ^3 trend.

Cross-check. The two mechanisms predict correlated suppression of the power spectrum at the same scales. A joint measurement of μ_0 from Euclid and $\sigma_{\rm crit}$ from Rubin, combined with the $\beta_m = \Omega_m/2$ prediction already confirmed at 0.2σ by Planck 2018, would constitute a three-way consistency test of the virial partition closure condition at cosmological and galactic scales simultaneously.

7 Discussion

The missing satellites problem has been approached primarily as a problem of baryonic physics: which processes remove gas from satellites, suppress star formation, or disrupt dark matter halos after they form. The IAM framework approaches it differently. The question is not what prevents baryons from settling into subhalos, but what determines whether dark matter halos virialize in the first place.

The answer, within IAM, is thermodynamic. A halo virializes when both halves of the virial partition can close. The geometric half always finds its channel in spacetime curvature. The informational half requires an available encoding surface. At the satellite mass scale, that surface must be a local black hole horizon. The minimum mass for black hole formation in a collapsing halo determines the minimum mass for virialization. Below that mass, the partition cannot close. The halo disperses. No satellite forms.

The open problem — the normalization offset of $\sim$100$\times$ in Equation 11 — is identified precisely. It requires the formal derivation of the local-to-global encoding rate criterion: the condition under which decoherence events register on the local black hole horizon versus the global cosmic horizon. This criterion involves the ratio of local to global decoherence rates and the thermodynamic temperature differential between the two horizons. Both quantities are calculable within IAM without additional free parameters [Mahaffey, 2026a]. The problem is identified here as an open problem rather than a solved one.

The scaling predictions — σ^3 for Mechanism B, σ^2 for cusp-core, σ^4 for M–σ — form a coherent family. All three derive from the same thermodynamic identity [Mahaffey, 2026g] at different boundary conditions. All three have the same coupling constant $\beta_m = \Omega_m/2$. The different power-law indices reflect different physical regimes: the minimum mass for virialization (Mechanism B), the encoding radius within a virialized halo (cusp-core), and the steady-state SMBH mass accumulated over a Hubble time (M–σ). A complete test of the virial partition closure condition requires consistency across all three scalings simultaneously.

Acknowledgments

The thermodynamic framework of T. Jacobson [Jacobson, 1995] is the foundation upon which this work is built. Timestamped predictions for the missing satellites problem and

cusp-core scaling were established in the GitHub repository on March 16, 2026, prior to the analysis presented here.

Data Availability

All MCMC chains, analysis scripts, and timestamped prediction figures are publicly archived at https://github.com/hmahaffeyges/IAM-Validation with canonical Zenodo DOI 10.5281/zenodo.18702042.

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